



NET PROFIT KILLS THE INTENTION

Why conventional investment models fail artisanal and small-scale gold mining in Africa

Investment economics, mercury reduction and the case for shared mining infrastructure

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1. Executive Summary

For decades, governments, development organisations, researchers and investors have sought to improve artisanal and small-scale gold mining (ASGM) in Africa. The objectives are familiar: reduce or eliminate mercury, improve recovery, formalise operations, introduce safer technology and create sustainable local economic development.

The evidence reviewed for this report suggests that the principal obstacle is not simply a lack of technology. It is also a problem of investment structure. Conventional investors commonly evaluate a small mine as a stand-alone asset: ore grade, tonnes, recovery, revenue, OPEX, CAPEX, net profit and return on investment. When the full cost of professional infrastructure is allocated to one small licence, the project may appear too small or too risky to finance.

The miner, however, does not stop mining because formal capital declines the investment. Production continues with the methods that are accessible and affordable. In many African ASGM areas, those methods still include mercury amalgamation.

The central proposition of this report is therefore that the unit of investment may be wrong. Instead of asking whether each small mine can finance a complete industrial system, investors should examine whether shared professional infrastructure can serve a portfolio of licensed mines. The economic question then moves from mine economics to system economics.

2. Background and Context

ASGM is economically significant across sub-Saharan Africa, but mercury remains deeply embedded in gold recovery. A 2024 review in the *Journal of Cleaner Production* synthesised 167 relevant publications from 1995-2024 and concluded that mercury, often combined with cyanide, remains a major source of environmental and health harm. The same review found that adoption of mercury-free technology depends not only on technical performance but on accessibility and the technology's ability to increase miners' incomes.

This matters for investment. If cleaner processing requires capital, but each small operation is judged too small to justify that capital, the sector becomes trapped: the mines most in need of improved technology are often the least able to finance it.

The problem is therefore not only environmental. It is also a capital-allocation and business-model problem.

3. Scope and Method

This report is a focused literature review and strategic analysis. It draws on peer-reviewed research concerning mercury management, mercury-free processing, ASGM processing centres, direct smelting and the economic organisation of small-scale mining, with particular attention to Tanzania, Ghana, Uganda and Burkina Faso.

The review is not a full systematic meta-analysis. Recovery results from individual field and laboratory studies are mineralogy- and process-specific and must not be treated as universal performance guarantees. The purpose is to identify recurring evidence relevant to the design and financing of shared mining infrastructure.

4. Findings

- 4.1 The research base is large, but solution evidence is much smaller
Kohio et al. (2024) reviewed 167 relevant publications on pollution and sustainability in sub-Saharan African ASGM. By contrast, a 2018 systematic review focused specifically on mercury-management interventions found only 22 intervention studies worldwide for 2007-2017. The authors concluded

that mercury reduction requires a comprehensive approach combining technology, education, mineralogy, economic considerations and government support.

This distinction is important: the damage caused by mercury is extensively documented; sustainable implementation of alternatives is much less well established.

4.2 Tanzania: technology exists, but adoption and access remain constraints

Maganga et al. (2023) reported that Tanzanian ASGM still uses inefficient and environmentally harmful practices including amalgamation. Their proposed mercury-free flowsheet uses comminution, two-stage gravity concentration and controlled recovery from concentrates and tailings.

The same study reported limited field uptake of retorts and borax smelting despite earlier demonstrations. It also described mineral demonstration centres at Katente, Lwamgasa and Itumbi. The centres provide training and processing services, but capacity and affordability remain material constraints.

Kinyondo and Huggins (2020) reached a related institutional conclusion. Tanzania established four training, processing and demonstration centres in 2019, but the authors found that the success of such centres depends heavily on affordability, flexibility and genuine demand for services.

In 2024, Maganga et al. tested direct smelting of Knelson gravity concentrates from the Nholi mine in Tanzania. Under low-sulfide conditions ($\leq 6.7\%$ sulfide), direct smelting achieved more than 87% gold recovery. The study demonstrates that a mercury-free gravity-to-concentrate route can be technically viable under suitable mineralogical conditions.

4.3 Ghana and Uganda: mercury-free processing can be competitive

Amankwah et al. (2010) compared direct smelting with amalgamation in Ghana. Laboratory direct smelting achieved 99.8% recovery versus 97% for amalgamation. In field tests, direct smelting averaged 98.3% compared with 88% for amalgamation.

A pragmatic field comparison in Tiira, Uganda, found that amalgamation recovered 1.0 g of gold in 53 minutes while using 4 g of mercury. Direct smelting recovered 1.4 g in 62 minutes without mercury - a 40% higher gold yield in that single comparison. The authors explicitly cautioned that the experiment was conducted only once, so the result should be treated as evidence of potential rather than a universal recovery advantage.

4.4 Burkina Faso: mercury is also part of the financing system

Brugger et al. (2022) demonstrated that mercury can perform a non-technical role in the ASGM value chain. Informal financiers prefinance extraction and provide mercury as part of arrangements that secure access to recovered gold and repayment. The authors describe mercury as a commitment and control device.

This finding is central to the investment argument. Removing mercury without replacing the surrounding financing and market structure may leave the underlying economic dependency untouched.

5. Analysis and Implications

The conventional investment calculation is not wrong; it may simply be applied at the wrong level.

A single PML or small mining site may have limited throughput, incomplete geological information, variable grades and high fixed costs relative to production. If one mine is expected to carry the full cost of laboratories, geology, technical staff, processing equipment, environmental management, maintenance, security, Gold Room functions and administration, the project can quickly fail a conventional net-profit or ROI threshold.

But the mine does not disappear when the spreadsheet says no. Mining continues, often through informal capital and lower-cost processing practices.

Shared infrastructure changes the unit of analysis. Multiple licensed operations can potentially share laboratories, technical specialists, training, maintenance, processing capacity, secure gold handling and management. Fixed costs can be distributed across a wider production base; equipment utilisation can increase; specialist expertise can serve several operations; and geological and operational risk can be diversified across a portfolio.

The investor's question then changes from:
'What is the net profit of this mine?'

to:

'What is the economic performance of the system serving these mines?'

This is the core meaning of the title Net Profit Kills the Intention. Net profit itself is not the problem. The problem arises when a conventional stand-alone mine calculation is used to reject a sector whose economics may depend on aggregation and shared infrastructure.

The research on mercury reinforces this point. Technical alternatives exist, but adoption repeatedly depends on affordability, incentives, access to capital, operational fit and market structure. Cleaner technology therefore needs a sustainable business model around it.

6. Recommendations

1. Evaluate portfolios, not isolated sites. Investment analysis should test aggregated production and shared infrastructure across multiple licensed operations rather than requiring every small mine to finance a complete processing organisation.
2. Separate mine economics from system economics. Site-level production performance remains essential, but shared technical, processing, training and administrative functions should be modelled at system level.
3. Make mercury reduction economically superior, not merely environmentally preferable. A replacement process must compete on recovery, speed, reliability, transparency, cash flow and access to market.
4. Use mineralogy before prescribing technology. The Tanzania direct-smelting study demonstrates that sulfide content materially affects performance. Mercury-free processing must therefore begin with ore characterization and test work
5. Design processing centres around demand, affordability and flexibility. Tanzania's experience shows that infrastructure alone does not guarantee utilisation.
6. Integrate financing and market access. Burkina Faso demonstrates that mercury can be embedded in informal financing relationships. A sustainable transition must offer miners an alternative route to working capital and transparent gold sales.
7. Position the NORSE/NMC concept as shared infrastructure rather than a technology sale. The strategic opportunity is to provide access to professional capability across a network of licensed operations, while retaining site-specific technical evaluation.

7. Risks and Limitations

The evidence does not support a claim that one mercury-free flowsheet will work for every ore body. Direct smelting results depend on concentrate grade and mineralogy, particularly sulfide content. Individual recovery figures cited in this report are study-specific.

Shared infrastructure also introduces concentration and governance risks: capacity bottlenecks, transport logistics, allocation of processing slots, assay trust, gold security, ownership of tailings and transparent revenue allocation all require robust operating rules.

Finally, the report advances an investment thesis rather than proving a complete financial model. The next analytical step should quantify the break-even point at which shared NMC infrastructure becomes economically superior to stand-alone processing across different numbers of sites, grades, throughputs and recovery scenarios.

8. Conclusion

Africa does not lack evidence that mercury is harmful, and it does not lack demonstrations of alternative processing methods. The recurring difficulty is converting technical possibility into a system that miners can access and investors can finance sustainably.

Conventional investment analysis tends to ask whether one small mine can support the cost structure of a complete modern mining operation. For many ASGM sites, the answer will be no. But that does not necessarily mean the sector is uninvestable.

The alternative is to aggregate. A network of smaller licensed producers can share professional infrastructure and create scale at system level without requiring each mine to become a miniature large-scale operation.

The strategic conclusion is simple:

Good intentions do not scale by themselves. Systems can.

Net profit kills the intention only when we calculate it in the wrong place.

9. References

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Appendix A - Supporting Material

Key evidence snapshot

- 167 publications reviewed in the 2024 sub-Saharan Africa pollution review.
- 22 intervention studies identified worldwide in the 2018 systematic review of mercury-management solutions.
- Tanzania (2024): >87% recovery from direct smelting under ≤6.7% sulfide conditions.
- Ghana (2010): field direct smelting 98.3% versus 88% amalgamation.
- Uganda (2019): 1.4 g direct smelting versus 1.0 g amalgamation in one pragmatic comparison; no mercury used in direct smelting.
- Tanzania's training/processing centres illustrate that affordability, flexibility and service demand are as important as equipment availability.
- Burkina Faso research shows that mercury can be embedded in informal prefinancing and control of gold marketing.

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